Small Group Research
http://sgr.sagepub.com/

# Development of Cognitive and Affective Trust in Teams: A Longitudinal Study

Sheila Simsarian Webber
*Small Group Research* 2008 39: 746
DOI: 10.1177/1046496408323569

**Small Group Research**
Volume 39 Number 6
December 2008 746-769

10.1177/1046496408323569
http://sgr.sagepub.com
hosted at
http://online.sagepub.com

# Development of Cognitive and Affective Trust in Teams

## A Longitudinal Study

Sheila Simsarian Webber
*Suffolk University*

The present research examines the development of two dimensions of trust, cognitive and affective, in student project teams over the course of a semester. Empirical examination of the evolution of multidimensional trust and its unique antecedents was explored. The results show that early trust emerges as a one-dimensional factor early in the life span of a team; cognitive and affective trust emerge as separate components over time; unique and distinct predictors positively and negatively affect early trust, cognitive trust, and affective trust; and affective trust has a stronger positive relationship with team performance than cognitive trust.

***Keywords:*** *trust; affective trust; cognitive trust; teams; longitudinal*

Interpersonal trust has an important history in the psychology, sociology, negotiation, strategy, and organizational-behavior disciplines (Cook & Wall, 1980; Currall & Judge, 1995; Deutsch, 1958; Lewicki & Bunker, 1995; Lewis & Weingert, 1985; Polzner, Crisp, Jarvenpaa, & Kim, 2006; Ring & Van de Ven, 1992; Zaheer, McEvily, & Perrone, 1998). More recently, trust has been identified as an important component of teamwork. Researchers have proposed the critical role of trust in the development of effective work team processes and the successful performance of the team (Dirks, 1999; Kirkman, Rosen, Tesluck, & Gibson, 2006; Polzner et al., 2006). The value of trust has been demonstrated in a variety of important work relationships. However, typically, a one-dimensional definition and measure of trust following the research of Mayer, Davis, and Schoorman (1995) has been used. In fact, Dirks and Ferrin's (2002) meta-analysis of trust in leadership found that 94% of the studies in this area used a one-dimensional cognitive-based definition and measure of trust.

Examining the development and changing nature of trust over time in team and interpersonal engagements is missing in the trust research. In general, scholarly research on trust has taken a snapshot, static approach to the concept (Rousseau, Sitkin, Burt, & Camerer, 1998) in dyadic situations. In the team literature, team trust is typically examined early in the development of the work team or in work teams with a short life span (i.e., laboratory teams). Missing in this research is an empirical examination of how trust evolves over time and the implications for the effectiveness of the team. Taking a longitudinal approach to understanding trust by building on the work of organizational and team researchers as well as interpersonal trust researchers in the psychology and sociology disciplines, the present research examines the composition and antecedents of early trust as well as established trust (cognitive and affective trust) in student project teams over the course of a semester. These teams are similar to task forces frequently used in a variety of organizations because they work interdependently for an extended time to accomplish a common objective in a limited time span (Hackman, 2003). Empirical examination of the theoretical arguments proposed in the trust literature for the emergence of cognitive and affective trust driven by different antecedents was explored. I was interested in extending the theoretical arguments proposed in the trust literature by examining the following: (a) Early trust emerges as a one-dimensional factor early in the life span of a team; (b) cognitive trust and affective trust emerge as separate components over time; (c) unique and distinct predictors affect early trust, cognitive trust, and affective trust; and (d) affective trust will have a stronger significant impact on team performance compared to cognitive trust.

## Interpersonal Trust

A variety of researchers have proposed that trust is multidimensional, consisting of a competency-based trust and an emotional-based trust (Cook & Wall, 1980; Johnson & Grayson, 2005; Jones & George, 1998; Lewis & Weingert, 1985; McAllister, 1995; Shapiro, Sheppard, & Cheraskin, 1992; Webber & Klimoski, 2004). Cook and Wall (1980) defined these two dimensions of interpersonal trust as (a) “faith in the trustworthy intentions of others” and (b) “confidence in the ability of others, yielding ascriptions of capability and reliability” (p. 40). Multidimensional trust was also developed through the work of Cummings and Bromiley (1996) who identified three dimensions of trust; affective, cognitive, and intended behavior. Their affective and cognitive dimensions are similar to those proposed by Cook and Wall, whereas behavioral intentions are similar to monitoring behaviors used as an antecedent of trust in this research. However, much of the multidimensional trust research has been theoretical or measurement oriented with only a few empirical tests to validate these ideas.

Following the multidimensional trust research ideas, McAllister (1995) argued that trust should be defined as both affective and cognitive. Affective trust, he argued, is grounded in reciprocated interpersonal care and concern or emotional bonds, whereas cognitive trust is grounded in individual beliefs about peer reliability and dependability as well as competence. Similarly, Jones and George (1998) argued for conditional and unconditional trust. Parallel to the definitions of cognitive and affective trust, conditional trust is based on knowledge and positive expectations of the other person, and unconditional trust is based on positive affect and mutual identification. Based on the review of the literature, it is possible to argue that trust is multifaceted. In addition, based on the above research, two components of trust consistently emerge: a cognitive element based on reliability, dependability, and competence and an affective (emotional) component based on close interpersonal relationships.

Although conceptual evidence and a few empirical studies show the components of cognitive and affective trust, a large number of researchers propose and measure a one-dimensional construct (e.g., Krishnan & Martin, 2006; Langfred, 2007; Mayer & Davis, 1999; Mayer & Gavin, 2005). For example, working from the framework of Mayer et al. (1995), Mayer and Davis (1999) conceptualized trust as one-dimensional. They measured trust in management using four items but found the measure had low reliability ($\alpha = .60$). A number of other researchers have adopted Mayer and Davis's (1999) original one-dimensional measure used in a working paper (Shoorman, Mayer, & Davis, 1996) to measure trust that also showed unreliability (e.g., Chiaburu & Baker, 2006; Ferrin, Kim, Cooper, & Dirks, 2007; Jarvenpaa & Leidner, 1999; Kim, Dirks, Cooper, & Ferrin, 2006). These results demonstrate that this one-dimensional measure is an unreliable assessment of trust. However, this research also shows that trust has been conceptualized as a one-dimensional construct.

## Team Trust

Dominant in the research on trust is an emphasis on interpersonal trust and organizational trust. Little research is available that examines trust at the team level of analysis (Surva, Fuller, & Mayer, 2005). Prior theoretical work has offered team trust as a viable and useful construct conceptualized at the team level of analysis (Dirks, 1999; Mayerson, Weick, & Kramer, 1996). However, empirical research is needed that measures and analyzes trust at the team level.

Recently, Surva et al. (2005) conceptualized and measured trust at the team level of analysis; however, their measure assessed the team's trust in another team, not the team's trust in their own team. In addition, their measure treated trust as a one-dimensional construct. Similarly, research by Jarvenpaa and Leidner (1999) used an adaptation of Schoorman et al.'s (1996) measure of trust, which is one-dimensional. They had a very small sample (21 teams) and did not report the reliability of the measure. In another article, research by Polzner, Crisp, Jarvenpaa, and Kim (2006) examined trust in geographically dispersed teams also using a one-dimensional measure. Finally, Prichard and Ashleigh (2007) showed that team building affects team trust also using a one-dimensional measure of team trust.

Research by Dirks (1999), however, measured trust at the team level of analysis by adapting the multidimensional measure by McAllister (1995) to the team level. This research was done in a laboratory setting using a short team task (tower building with wooden blocks). The result of having a short time frame for the teams was that all items loaded on one factor rather than confirming the multidimensional nature of the construct that has been found to emerge in longer working relationships. Subsequent research by Ferrin and Dirks (2003) has studied trust in teams also using a laboratory experiment (wilderness survival) and a one-dimensional measure of trust. As discussed below, early trust in short-term teams could explain the lack of multidimensionality obtained in this research. The present research adopts a different approach by measuring trust over time in student project teams that work together for 3 months to accomplish a common goal.

## Dynamics of Cognitive and Affective Trust

The theoretical arguments regarding the development of cognitive and affective trust, as well as conditional and unconditional trust, have proposed that the two dimensions of trust develop and emerge over time (Jones & George, 1998; McAllister, 1995). According to Kim, Ferrin, Cooper, and Dirks (2004), trust develops over time; however, early high levels of trust are possible and may be driven by cognitive cues from group membership and reputation. Affective trust has been thought to develop later in the life of an interpersonal relationship (Williams, 2001).

I argue that a basic one-dimensional level of trust needs to be present before a two-dimensional trusting relationship with both cognitive and affective components can develop. Research has yet to demonstrate that these two components of trust do differentiate themselves over time. Research on the cognitive and affective components of trust in teams and interpersonal relationships often fails to find a two-dimensional construct (Dirks, 1999) and has not explored the development of trust over time. The study of trust in interpersonal and team relationships will be advanced by identifying the nature of trust at different points in the life of a team. Such knowledge can then be used to more precisely determine the factors that facilitate and hinder trust at specific times in the interpersonal relationship.

The present research examines trust at two different points in time, early in the life of the team and at the end of the team's lifespan. I propose that over time, the two dimensions of trust will emerge such that early trust will be one dimension primarily guided by early reactions of general competence and likability. Established trust, measured at the end of the team's life span, will be two dimensions distinguished as cognitive and affective trust, and the two dimensions will be related but distinct components of trust.

*Hypothesis 1:* Early trust will be one-dimensional.

*Hypothesis 2:* Trust measured at the end of the team's life span, "established" trust, will be two-dimensional, and cognitive and affective trust will be related but distinct components.

## Escalation of Early Trust and Cognitive and Affective Trust

Researchers agree that individual ability or competency is a strong predictor of trust (Mayer et al., 1995; McAllister, 1995). However, in the early stages of the life of a team, ability is difficult to assess without some prior familiarity with the individuals of the team. Therefore, the early formation of trust is proposed by McKnight, Cummings, and Chervany (1998) to be affected by categorization processes based on early knowledge. Such early knowledge about other team members could be obtained from prior familiarity with the team member or knowledge about the team member's reputation. In the project team context, early knowledge of other team members could be easily gained from any prior experiences working together and/or from other people's prior experiences with the team member. Therefore, the following hypothesis is proposed:

*Hypothesis 3:* Familiarity with team members will be positively related to early trust.

Differentiating the antecedents of cognitive and affective trust in established interpersonal relationships resulted in McAllister (1995) proposing that reliable role performance would be positively associated with cognitive trust and that citizenship behavior and interaction frequency would be positively associated with affective trust. Interaction frequency is described as the frequency of interaction between two individuals. Citizenship behaviors include behaviors that are personally chosen rather than role-prescribed, serve legitimate needs, and demonstrate interpersonal care and concern. These behaviors correspond to organizational citizenship behavior (Organ, 1988; Smith, Organ, & Near, 1983), which is defined as behavior intended to provide help and assistance, but outside the individual's work role and not directly rewarded. Reliable role behavior is described as success of past role-related duties. Consistent with his prediction, McAllister found that interaction frequency and the demonstration of citizenship behaviors led to affective trust. He uncovered less support for the impact of reliable role behavior on cognitive trust. However, building on McAllister's research, Johnson and Grayson (2005) found that service delivery performance is positively related to cognitive trust and not related to affective trust.

The research examining interpersonal trust demonstrates that different behaviors affect each aspect of trust offering additional support for the two components of trust. However, to the author's knowledge, no research exists of a longitudinal nature examining the impact of such behaviors on cognitive and affective trust after controlling for prior levels of trust. This additional control of prior trust is critical for determining the stability of early trust as research has proposed that early trust has a positive impact on trust developed later in interpersonal or team relationships (McKnight et al., 1998). In addition, by controlling for early trust, it is possible to demonstrate the primary behavioral antecedents of established trust.

Interacting frequently with the members of a team is considered an important element of team effectiveness (Ashleigh & Nandakhumar, 2007; Hackman, 2003). When teams are communicating, they are sharing important information about the nature of the team task, the team's progress in completing the task, and any challenges the team is facing. Frequent interaction also enables team members to build a relationship with the other members of the team. In other words, teams with frequent interaction are more likely to develop a strong relationship and rapport among the team members, resulting in greater affective trust.

Citizenship behaviors, on the other hand, involve team members helping other team members and the team as a whole to be successful. Going above and beyond to ensure that the team is a success is a key part of citizenship behaviors (Organ, 1988). Typically, citizenship behaviors are considered extra role behaviors and are not specifically part of an individual's tasks in a team environment. However, performing extra role behaviors such as helping other team members or working longer hours to ensure a successful project facilitate a stronger relationship and emotional bond among the members of the team. Organizational citizenship behavior is similar to benevolence, which has been linked to trust in organizations (Mayer et al., 1995). This type of relationship should result in greater affective trust or trust based on care and concern for the team and its members.

Controlling for familiarity and early trust, the research question remaining is (a) if interaction frequency and citizenship behaviors will be positively related to affective trust and (b) that these behaviors will not be related to cognitive trust.

*Hypothesis 4a:* Interaction frequency will be positively and significantly related to affective trust after controlling familiarity, and early trust and interaction frequency will not be positively and significantly related to cognitive trust.

*Hypothesis 4b:* Citizenship behaviors will be positively and significantly related to affective trust after controlling for familiarity, and early trust and citizenship behaviors will not be positively and significantly related to cognitive trust.

Second, after controlling for familiarity and early trust, the research question is whether certain behaviors have an impact on determining cognitive trust. Cognitive trust is typically developed through demonstrations of competence, reliability, and dependability. In other words, teams that reliably perform their tasks such that they are done on time and according to the requirements will have more cognitive trust. Reliable performance also involves behaviors that demonstrate competence and dependability. These behaviors are proposed to be positively related to cognitive trust, after controlling for familiarity and early trust, and will not be related to affective trust.

*Hypothesis 5:* Reliable performance will be positively and significantly related to cognitive trust after controlling for familiarity and early trust, and reliable performance will not be positively and significantly related to affective trust.

## Deterioration of Cognitive and Affective Trust

Taking a different approach to understanding trust, a group of researchers has examined the impact of behaviors or actions that operate as substitutes for trust (i.e., formal mechanisms; Sitkin & Roth, 1993; Zucker, 1986). However, research in this area has suggested that formal mechanisms such as contracts can actually reduce trust, rather than serve to effectively remedy trust. In fact, in some cases, such behavior/actions can result in greater formality or rigid procedures and policies along with increasing the distance and coldness between the individuals (Zucker, 1986). The present research extends earlier work on interpersonal trust and proposes that monitoring behaviors by team members become formal mechanisms that negatively affect the development of cognitive and affective trust in teams.

Monitoring behavior includes tracking the work of others, creating backup plans, and working around team members to get tasks done. Specifically, I propose the following hypothesis:

*Hypothesis 6:* Monitoring behaviors will be negatively and significantly related to cognitive and affective trust after controlling for familiarity and early trust.

## Interaction of Early Trust and Behaviors Affecting Cognitive and Affective Trust

Prior early trust along with the team behaviors described above (i.e., reliable performance, interaction frequency, citizenship behavior, and monitoring) are thought to positively affect cognitive and affective trust. In addition, the prior early trust will moderate the relationship between team behaviors and the development of cognitive and affective trust. In other words, having a trusting foundation within the team will facilitate the positive impact of reliable team performance behaviors on cognitive trust and will enhance the impact of interaction frequency and citizenship behavior on affective trust. Therefore, the following interactions are offered:

*Hypothesis 7a:* The relationship between reliable performance and cognitive trust is moderated by early trust such that reliable performance will be more strongly related to cognitive trust when early trust is high.

*Hypothesis 7b:* The relationship between interaction frequency and affective trust will be moderated by early trust such that interaction frequency will be more strongly related to affective trust when early trust is high.

*Hypothesis 7c:* The relationship between citizenship behaviors and affective trust will be moderated by early trust such that citizenship behaviors will be more strongly related to affective trust when early trust is high.

## Cognitive and Affective Trust and Team Performance

Trust has consistently been linked to effective performance (Dirks & Ferrin, 2002; Mayer et al., 1995; McAllister, 1995). However, differentiating between the impact of cognitive and affective trust on team performance has yet to be done. Theoretically, cognitive trust has been argued to be more challenging to sustain once it is developed compared to affective trust (McAllister, 1995). Specifically, a team could have high levels of cognitive trust then experience a problem completing a task on time, resulting in a decrease in cognitive trust.

On the other hand, affective trust is thought to persist even under these types of situations (McAllister, 1995). Affective trust is argued to withstand challenges in the interpersonal relationship and requires more persistent emotional problems to decrease (Williams, 2001). Therefore, researchers have argued that affective trust is particularly important for interpersonal relationships because once it is developed, it persists in the long term, something not achieved by cognitive trust. Furthermore, achieving affective trust allows for short-term behavioral problems to occur and be forgiven (Jones & George, 1998; McAllister, 1995). In addition, affective trust is driven by behaviors that extend beyond task-related competency to include extra role behaviors delivered during the life of the team. Therefore, affective trust, once it is developed in a relationship, should be a more powerful and longer term type of trust (McAllister, 1995; Williams, 2001). The performance of a work team should be positively affected by both cognitive and affective trust; however, affective trust, because of its deep-rooted nature should have a more positive relationship with team performance.

*Hypothesis 8:* Cognitive and affective trust will have a positive significant relationship with team performance; however, the relationship with affective trust will be stronger.

# Method

## Participants

Participants were recruited from Organizational Behavior classes in a large Canadian university. The author approached class instructors and asked for their class participation. The class was asked to form teams of three ($N = 9$) to four ($N = 69$) students to complete the same team project. The same undergraduate classes (i.e., Organizational Behavior) were used for two semesters. Nine professors were approached to participate, and four agreed including the author, resulting in nine classes with an average of 40 students in each class. The teams worked over the course of the semester to complete a team tutorial project designed by the author. Team tutorial topics were assigned by the author and included tasks such as teaching the class how to successfully manage a cross-functional team or how to complete a performance appraisal. Each team in a class was given a different topic. Teams were required to research the topic area and prepare a how-to presentation and paper on their assigned topic. A total of 378 students participated at Time 1 resulting in 106 teams, 338 of the same students participated at

Time 2 resulting in 98 teams, and a total of 294 of the same students participated at Time 3 resulting in a total of 78 teams. The 78 teams resulted in data for Time 1, Time 2, and Time 3 and were therefore used for all the analyses. All teams had either three or four members, and all team members needed to complete the survey at each time to be included in the final 78 teams for the analysis.

## Team Instructions

Teams were given the following instructions for completing the team tutorial:

> The objective of the team tutorial sessions is to supplement the text with "need-to-know" management activities. Ten teams of four students will be formed. Each team will be given one tutorial topic. Teams can use any number of approaches (interviews, books, articles, Internet, magazines, etc.) to prepare for the tutorial. Each team needs to submit an outline of their team tutorial in the middle of the semester (please see course outline for the date) and a *five-page* summary of their tutorial when they make their presentation. The presentations will be for *10 minutes* per team. Teams need to monitor their own time to complete the presentation within the 10-minute time slot. Individually, each team member needs to submit a *one-page* self- and peer evaluation. You are to assess your contribution to your team and each of your team members' contributions in paragraph format. You are also asked to evaluate yourself and your team members by giving each either the same grade as the team or a grade lower than the team grade. Your evaluations will be 25% of the grade each team member receives on the team tutorial. The other 75% of your grade will be the grade given to the team for the tutorial.

## Procedure

The author and a research assistant attended each class at the 2nd week of the semester to inform the students about the research and obtain their consent to participate in the research. The author also provided a detailed introduction about the team tutorial project and answered any questions from the students. Students formed their teams during this class and were given the first survey. Students were given 15 minutes at the end of class to complete the survey and return it to the author. Three weeks later, the student teams turned in an outline of their proposed project to their professor. The author or the research assistant returned to each class at this time and gave the students the second survey to complete. Again, the students completed the survey during class time. Five weeks later (Week 10 of the semester), the student teams gave a 10-minute presentation and turned in a team paper. The author or the research assistant returned at this time to each class and gave the students the third survey to complete during class time.

## Measures

*Familiarity*. Familiarity was assessed at Time 1 by adapting a measure used by Webber, Chen, Marsh, and Payne (1999) to the student project team context. The measure consisted of four questions that asked the team members to evaluate their prior familiarity with each of their team members (e.g., "How familiar are you with the strengths and weaknesses of this team member?" using 1 (*I am not familiar*) to 5 (*I am very familiar*) with a reliability of .84. Average scores were then generated across team members to obtain a team score. See the appendix for the list of the items.

*Team trust*. Team trust was measured at Time 2 and Time 3 by using a measure of interpersonal trust by McAllister (1995) taken to the team level of analysis. The same measure was used by Dirks (1999) to measure team trust. The Time 2 measure was taken after the team had been formed for 3 weeks, and the Time 3 measure was taken at the end of semester after the team had been working together for 8 weeks. Trust was not measured at Time 1 when the teams were first formed as many of the team members had no information about their team and its members to evaluate their trust. Scanzoni (1979) as well as Rempel, Holmes, and Zanna (1985) argued that trust is not likely to appear at the beginning of a relationship that has no past experience. Three items assessed cognitive trust, and three items assessed affective trust. Factor analysis using principal components with varimax rotation was used at Time 2 and demonstrated a one-factor solution with an eigenvalue of 2.66 with Items 1, 2, 3, 5, and 6. Confirmatory factor analysis demonstrated a good fit for a one-factor solution, $\chi^2(5, N = 276) = 27.98$, $p < .00$, Comparative Fit Index (CFI) = .97, Normed Fit Index (NFI) = .97, and root mean square error of approximation (RMSEA) = .14. At Time 3, factor analysis demonstrated a two-factor solution with eigenvalues of 2.21 and 2.18. Confirmatory factor analysis showed that a two-factor solution was a good fit, $\chi^2(8, N = 276) = 30.99$, $p < .00$, CFI = .98, NFI = .98, and RMSEA = .11. The factor structure showed affective trust as a distinct component of trust, and the relationship between cognitive and affective trust was $r = .64$, $p < .05$. For Time 3, cognitive trust was assessed with Items 1, 2, and 3 and had a reliability of .84. Affective trust was assessed with Items 4, 5, and 6 and had a reliability of .88. See Table 1 for results of the two factor analyses.

**Table 1**
**Factor Analysis Results—Early Trust and Established Trust**

| | | | Established Trust | |
|---|---|---|---|---|
| Item | Early Trust | | Cognitive | Affective |
| The members of my team approach the team project with professionalism and dedication. | .72 | | .82 | |
| Given the track record of my team members, I see no reason to doubt their competence and preparation for the upcoming presentation. | .80 | | .83 | |
| I can rely on the members of my team not to make my job more difficult by careless work. | .69 | | .78 | |
| We (the team) have a sharing relationship. We can openly share our ideas and feelings. | .10 | .96 | | .86 |
| We can talk freely to each other about difficulties we are having in completing the project and know that each other will listen. | .64 | | | .86 |
| If I shared my ideas and project-related problems with the members of my team, I know they would respond constructively and caringly. | .78 | | .39 | .72 |

*Interaction frequency*. Interaction frequency was assessed at Time 2 by adapting a measure of interaction frequency from McAllister (1995) to the team level of analysis. Three questions were asked about the frequency of interaction (e.g., "How frequently do members of your team initiate team-related interactions with you?" using a scale of 1 (*once every 2 months*) to 5 (*daily*). The scale reliability was .80.

*Citizenship behaviors*. Citizenship behaviors were assessed at Time 2 by adopting a measure of peer affiliative citizenship behavior from McAllister (1995) to the team level of analysis. Seven items assessed behaviors that are personally chosen and demonstrate care and concern (e.g., "My team members willingly help each other, even at some cost to personal productivity"). Items were assessed using a scale from 1 (*strongly disagree*) to 5 (*strongly agree*). Scale reliability was .85.

*Reliable performance*. Reliable performance was assessed at Time 2 by adopting a measure of reliable performance from McAllister (1995) to the team level of analysis. Three items were used to measure the team's performance (e.g., "My team performs all tasks that are expected of them") on a scale from 1 (*never*) to 5 (*always*). Scale reliability was .91.

*Monitoring behavior*. Monitoring behavior was assessed at Time 2 by adopting a measure by McAllister (1995) to the team level of analysis. Four items were measured addressing the team's need to monitor team members (e.g., "I have sometimes found it necessary to work around team members to get things done the way that I would like them done") and assessed using a scale from 1 (*strongly disagree*) to 5 (*strongly agree*). Scale reliability was .85. The appendix lists the items for familiarity, interaction frequency, citizenship behaviors, reliable performance, and monitoring.

*Team performance*. Team performance was assessed by the author for each class. All team projects were graded by the author on the basis of the same criteria and were given a grade from 0 to 100 points. The average team grade was 79. There were 54 teams with performance information. Final project scores could not be obtained for 24 of the teams in the sample because of one professor was not willing to have the author make copies of the papers to evaluate for this research.

*Data aggregation*. The unit of analysis for the hypotheses is the work team. For the 78 teams, within-team and between-team analysis was conducted on each of the variables tested. The results of this analysis showed that for all variables, the mean $R_{wg(j)}$ was .80 or greater ($R_{wg(j)}$ = .79-.99) (see James, Demaree, & Wolf, 1984). In addition, intraclass correlations ICC(1) and ICC(2) were examined. The results show significant ICC(1) = .25-.47 and ICC(2) = .53-.67 values, $F$s = 1.38-2.23, $p < .05$. ICC(1) demonstrates variability between groups, and ICC(2) shows reliability of the group mean. Based on these results and because of the research questions and hypotheses being drawn at the team level, the data were aggregated to the team level for the remaining analyses. Individual responses to the items composing the variables were averaged across all team members. Table 2 shows the means, standard deviations, and correlations at the team level of analysis.

## Results

The results of the factor analysis demonstrate support for Hypotheses 1 and 2. At Time 2, trust was found to be one-dimensional. At Time 3, affective

**Table 2**
**Means, Standard Deviations, and Correlations**

| Variable | *M* | *SD* | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1. Familiarity (Time 1) | 1.89 | .82 | | | | | | | | |
| 2. Early trust (Time 2) | 4.00 | .45 | .24* | | | | | | | |
| 3. Cognitive trust (Time 3) | 4.00 | .51 | .16 | .13 | | | | | | |
| 4. Affective trust (Time 3) | 4.14 | .50 | .17 | .15 | .64** | | | | | |
| 5. Reliable performance (Time 2) | 4.35 | .66 | .19† | .43** | .06 | .21 | | | | |
| 6. Citizenship behavior (Time 2) | 3.77 | .51 | .37** | .77** | .14 | .23* | .44** | | | |
| 7. Interaction frequency (Time 2) | 3.63 | .41 | .28* | .37** | .03 | .09 | .43** | .43** | | |
| 8. Monitoring (Time 2) | 2.78 | .52 | –.05 | .05 | –.30** | –.17 | –.13 | –.34** | –.24* | |
| 9. Team Performance | 79.06 | 8.43 | –.01 | .15 | .22† | .27* | .01 | .09 | –.06 | –.11 |

*$p \leq .05$. **$p \leq .01$. †$p < .10$.

trust clearly emerges as a distinct component of trust, and the relationship between cognitive and affective trust is $r = .64$, $p < .05$. See Table 1 for the results of the two factor analyses. These findings are the first, to the author's knowledge, that demonstrate empirically that the two components of trust emerge over time. Furthermore, the results show that teams with a short life span may never have the time to distinguish between these two components.

Hypothesis 3 proposed that familiarity will be positively and significantly related to early trust. The results show support for this hypothesis ($R^2 = .06$, $\beta = .24$, $p = .03$) such that the more familiar the team members are, the greater the early trust within the team. Hypothesis 4a proposed that interaction frequency would be positively and significantly related to affective trust after controlling for familiarity and early trust. The results show that familiarity ($R^2 = .03$, $\beta = .17$, $p = .14$) and early trust ($R^2 = .02$, $\beta = .15$, $p = .19$) are not significantly related to affective trust; therefore, because of the lack of relationship and the sample size, they were not controlled in the analyses. No support for the relationship between interaction frequency and affective trust was found ($R^2 = .01$, $\beta = .09$, $p = .46$). Interaction frequency does not measure the nature of the interaction that may explain the lack of relationship. The results also show that interaction frequency is not related to cognitive trust ($R^2 = .00$, $\beta = .03$, $p = .79$). Hypothesis 4b proposed that citizenship behavior, measured at Time 2, would be significantly and positively related to affective trust, measured at Time 3. The results show support for this hypothesis ($R^2 = .05$, $\beta = .23$, $p = .04$) such that the more citizenship behaviors demonstrated by team members at Time 2, the greater the affective trust at Time 3. Citizenship behavior is not related to cognitive trust ($R^2$ = .02, β = .14, $p$ = .23).

Hypothesis 5 proposed that reliable project performance, measured at Time 2, is significantly and positively related to cognitive trust measured at Time 3 (the end of the teams' life span) after controlling for familiarity and early trust. The results show no relationship between cognitive trust measured at Time 3 and familiarity ($R^2$ = .03, β = .16, $p$ = .17) and early trust ($R^2$ = .02, β = .13, $p$ = .26); therefore, they were not controlled in the analyses. The results show no support for Hypothesis 4 ($R^2$ = .00, β = .06, $p$ = .60).

Hypothesis 6 proposed that monitoring behavior would be negatively and significantly related to cognitive and affective trust after controlling for early trust and familiarity. Because of the lack of significant relationships between cognitive and affective trust and early trust and familiarity, these variables were not controlled for in the analyses. The results show some support for this hypothesis such that monitoring behavior is significantly and negatively related to cognitive trust ($R^2$ = .08, β = –.30, $p$ = .01), but not affective trust ($R^2$ = .03, β = –.17, $p$ = .13). Table 3 shows the results of the regression analyses.

The results show support for Hypothesis 7a such that early trust does moderate the relationship between reliable performance and cognitive trust. Specifically, when early trust and reliable performance are both high, cognitive trust is significantly higher ($\Delta R^2$ = .06, β = 3.36, $p$ = .04) (Figure 1). No significant interactions were found for early trust moderating the relationships between interaction frequency and citizenship behaviors with affective trust.

Hypothesis 8 proposed a positive relationship between cognitive and affective trust and team performance with affective trust having a stronger relationship. The results show that cognitive trust is not significantly related to team performance ($R^2$ = .05, β = .22, $p$ = .10) and that affective trust is significantly and positively related to team performance ($R^2$ = .07, β = .27, $p$ = .05).

## Discussion

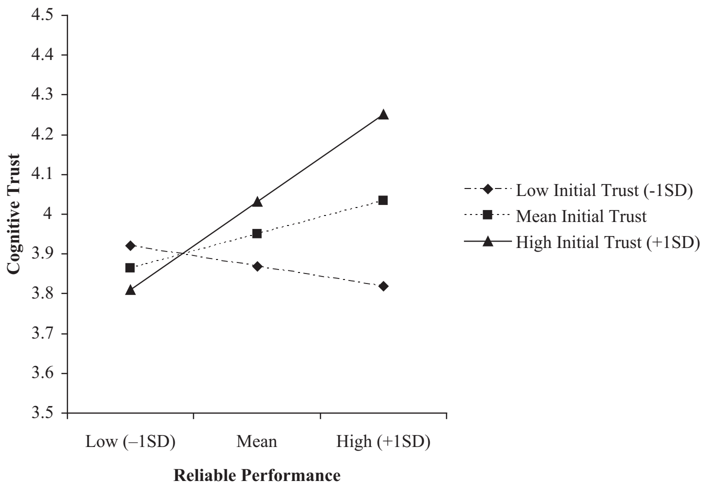
The present research extends the trust literature to examine four important propositions: (a) Early trust emerges as a one-dimensional factor early in the life span of a team; (b) cognitive and affective trust emerge as separate components over time; (c) unique and distinct predictors affect early trust, cognitive trust, and affective trust; and (d) cognitive and affective trust

**Table 3**
**Regression Results Hypotheses 5-8**

| Step and Variable | *B* | *SE B* | β | $R^2$ |
|---|---|---|---|---|
| Affective trust (Time 3) | | | | |
| Step 1: interaction frequency (Time 2) | .11 | .14 | .09 | .01 |
| Step 1: citizenship behavior (Time 2) | .23 | .11 | .23 | .05* |
| Step 1: monitoring behaviors (Time 2) | –.17 | .11 | –.17 | .03 |
| Cognitive trust (Time 3) | | | | |
| Step 1: reliable project performance (Time 2) | .05 | .09 | .06 | .00 |
| Step 1: monitoring behavior (Time 2) | –.30 | .12 | –.31 | .09 |
| Step 1: | | | | |
| Early trust (Time 2) | .14 | .14 | .13 | .02 |
| Reliable project performance (Time 2) | .01 | .10 | .01 | |
| Step 2: Early Trust × Reliable Performance | .46* | .22 | 3.36* | .07 |
| | | | | $\Delta R2 = .06$* |
| Team performance | | | | |
| Step 1: cognitive trust (Time 3) | 3.93 | 2.39 | .22 | .05† |
| Step 1: affective trust (Time 3) | 5.12 | 2.56 | .27 | .07* |

$^*p \leq .05$. $^{**}p \leq .01$. $^{\dagger}p < .10$.

**Figure 1**
**Early Trust Moderating the Relationship Between Reliable Performance and Cognitive Trust**



do differentially affect team performance. These propositions were tested with a longitudinal examination of trust in student project teams.

The results show that in Week 8 of working together as a team, affective and cognitive trust emerged as separate, yet related components. Prior to this, during early team formation, only one component of trust emerged. This finding highlights that those researchers examining trust in a static moment in time may be capturing only that particular phase of the team or interpersonal relationship and should consider interpreting their results with caution to the particular time period.

Second, I was particularly interested in the unique antecedents affecting each aspect of trust (early trust, cognitive trust, and affective trust). This research shows that early trust is developed through prior familiarity, consistent with the theoretical arguments by McKnight et al. (1998). However, familiarity does not significantly affect cognitive and affective trust later in the life of the team. In other words, familiarity only affects trust when other behaviors are not available (i.e., at the beginning of the life of the team) to determine trust within a team. Trust developed later in the project team is affected by its own unique antecedents driven by the actual behaviors of team members during the life of the team. Specifically, affective trust is developed through citizenship behaviors, which include doing extra things for team members, willingly helping each other, and taking a personal interest in the team.

Differing from affective trust, this research shows that cognitive trust is affected by the interaction of early trust and reliable performance such that in high early trust situations, the addition of reliable performance significantly increased cognitive trust. Therefore, having high early trust appears to be a necessary platform for reliable performance to affect cognitive trust later in the life of the team. Independently, neither early trust nor reliable performance significantly affected cognitive trust. The lack of a direct relationship between early trust and reliable performance demonstrates that particular conditions are necessary for these variables to affect cognitive trust.

Finally, monitoring behaviors were found to significantly and negatively affect the development of cognitive trust. Monitoring behaviors include diligently tracking the work progress of others, creating backup plans in anticipation of late work, and working around team members to get the tasks completed. Such behaviors resulted in a significant decrease in cognitive trust, but not affective trust. These results offer further evidence for the unique antecedents of cognitive and affective trust. Both cognitive trust and affective trust also were positively related to the performance of the team. Consistent with the theoretical ideas that affective trust is more durable and may be a more powerful form of trust in interpersonal and team relationships, this research shows that affective trust has a stronger significant relationship with team performance compared to cognitive trust. These results offer further evidence of the importance of trust and the distinct nature of cognitive and affective trust in team environments.

## Theoretical and Practical Implications

Theoretically, the findings show that researchers studying trust in interpersonal and team relationships should consider the life span of the relationships. Specifically, early in the life of a team, early trust is likely to emerge; however, later in the life of the team, both cognitive and affective trust will be established. Prior research has shown mixed support for the emergence of cognitive and affective trust (Dirks, 1999); however, this appears to be largely due to the life span of the team and the time of trust assessment. The present study demonstrates that there are two distinct forms of trust that emerge as a relationship develops; therefore, future theories and models should consider incorporating both the cognitive and the affective components of trust.

Second, the present research provides early empirical demonstration for the unique antecedents of early trust as well as cognitive and affective trust. These findings demonstrate that early trust is driven by previous history and familiarity, affective trust is driven by extra role behaviors, and cognitive trust is driven by in-role behaviors. Furthermore, the results show that monitoring behaviors have no impact on the decline of the affective component of trust but do reduce cognitive trust. These findings support the idea that affective trust is more enduring and plays a considerable role in the sustainability of interpersonal and team relationships in the workplace, particularly if an invalidation of the cognitive counterpart occurs. These findings are further substantiated by the significant positive relationship between affective trust and team performance.

In addition, this research finds that cognitive trust operates very differently. The findings show that early trust is a necessary condition for reliable performance to positively affect the development of cognitive trust. Without early trust, reliable performance has no relationship with cognitive trust. In addition, the present research shows that monitoring behaviors negatively affect the development of cognitive trust. Finally, cognitive trust does not have a significant relationship with performance. Trust researchers need to give particular attention to the development and impact of cognitive trust, particularly in long-term teams.

Practically, this research provides organizations, managers, and team leaders a better understanding of how trust develops over time. First, this research demonstrates that the two components of trust only emerge after a team has been working together for some time. The affective component of trust has been argued to be particularly important for cooperation and communication within teams (Williams, 2001). Therefore, in practice, organizations, managers, and team leaders could provide longer team engagements or accelerate the development of affective trust through team-building activities. Teams that never have the opportunity to distinguish affective from cognitive trust may have more difficulty successfully coordinating and communicating.

Second, organizations, managers, and team leaders should provide opportunities to help teams obtain early trust as it appears to be an important contributor to future trust. In addition, teams need the opportunity to progress from early trust to established trust quickly. Incorporating team-building activities into the team experience, as well as providing social gatherings to facilitate the formation of close relationships among the team members, may result in the quick development of affective trust. Research by Prichard and Ashleigh (2007) found that team training resulted in greater trust compared to teams with no training. Certainly, I would argue that those teams that form together, and need to perform to meet tight deadlines, are less likely to have the opportunity or the desire to obtain established trust.

## Limitations and Future Directions

First, the sample was undergraduate student teams from a large university. This limits the generalizability of the research to real teams in organizations. However, because the purpose of this research was to isolate the development of trust in team relationships, I believe the results offer important information regarding the development of trust in a longitudinal team. More research is necessary to further generalize these findings. Second, I did not examine the development of distrust in these work teams. Some researchers have argued that distrust is another component of trust in working relationships and should be considered along with the positive forms of trust, modeled in theories and examined empirically (Jones & George, 1998; Sitkin & Roth, 1993). In addition, this research approaches trust as a multidimensional construct focusing on two dimensions. Some researchers take a different perspective and believe that trust is a one-dimensional construct (Mayer & Davis, 1999) or a three-dimensional construct (Cummings

& Bromiley, 1996). This research shows support for the idea that a two-dimensional trust emerges over time.

Another limitation of this study was that students were able to select their own teams, which may have caused a self-selection bias for the research data. Similarly, each team typically had four members, which was a general requirement for the project. Having a set number of people for each team may not be realistic to how teams function in their natural organizational context. In addition, the performance of a team was assessed only by the author. Ideally, the team's performance would have been assessed by multiple people, and interrater reliability would have been calculated to ensure consistency across evaluations. A final limitation to this research is in the measurement of familiarity. It is possible that a team member could be familiar with another team member from a problematic prior class encounter. In other words, prior familiarity with someone could represent knowledge of a negative reputation, which would hinder the development of trust. This aspect of familiarity was not measured in the research. Similarly, the measure of interaction frequency does not capture the nature of the interactions. It is possible that teams had a lot of interaction but that it was not helpful for completing the task.

## Conclusion

In conclusion, this research demonstrates that trust has two components, one that is based on reliability, dependability, and competence, and a second component that is based on care, concern, and emotional bonds. I show that these two components of trust only emerge after the team has worked together for 8 weeks. In addition, I provide early evidence to the argument that early trust, cognitive trust, and affective trust are different aspects of trust that emerge at different points in time during interpersonal and team relationships. Researchers examining trust in interpersonal and team relationships should design and interpret their trust research with regard to the particular development time of the interpersonal or team relationship. Furthermore, researchers should include the two components of affective and cognitive trust in their research as they are distinct aspects of trust with unique antecedents and differing impacts on effectiveness. Researchers who continue to measure and theorize a one-dimensional trust are missing an important aspect of their research of interpersonal relationships.

## Appendix
### Items for Interaction Frequency, Citizenship Behavior, Reliable Performance, and Monitoring

| Item |
|---|
| Familiarity |
| How well do you know the academic reputation of this team member? |
| How well do you know this team member personally? |
| How well do you know the strengths and weaknesses of this team member? |
| How familiar are you with the way this team member works? |
| Citizenship behavior |
| My team members have taken a personal interest in the team. |
| My team members willingly help each other, even at some cost to personal productivity. |
| When making decisions in class that affect the team, my team members try to take each other's needs and feelings into account. |
| My team members frequently do extra things they know they will not be rewarded for, but which makes our work with the team more productive. |
| My team members take time to listen to each other's problems and worries. |
| My team members try not to make things more difficult for each other by their careless actions. |
| My team members pass on new information that is useful to the team. |
| Reliable performance |
| My team fulfills responsibilities specified in the project description. |
| My team performs all tasks that are expected of them. |
| My team meets formal performance requirements of the project. |
| Interaction frequency |
| How frequently do you initiate team-related interaction with members of your team? |
| How frequently do members of your team initiate team-related interaction with you? |
| How frequently does your team interact for project purposes? |
| Monitoring |
| I have sometimes found it necessary to work around team members to get things done the way that I would like them done. |
| I keep a close track of my interactions with team members, keeping track of instances when they do keep track of their end of the bargain. |
| The quality of work I receive from members of this team is only maintained by my diligent monitoring of members. |
| Rather than just depending on some team members to come through, I try to have a backup plan ready. |

## References

Ashleigh, M. J., & Nandakhumar, J. (2007). Trust and technologies: Implications for organizational work practices. *Decision Support Systems*, *43*, 607-617.

Chiaburu, D. S., & Baker, V. L. (2006). Extra-role behaviors challenging the status-quo: Validity and antecedents of taking charge behaviors. *Journal of Managerial Psychology*, *21*, 620-637.

Cook, J., & Wall, T. (1980). New work attitude measures of trust, organizational commitment, and personal need nonfulfillment. *Journal of Occupational Psychology*, *53*, 39-52.

Cummings, L. L., & Bromiley, P. (1996). The Organizational Trust Inventory (OTI): Development and validation. In R. M. Kramer & T. R. Tyler (Eds.), *Trust in organizations: Frontiers of theory and research* (pp. 68-89). Thousand Oaks, CA: Sage.

Currall, S. C., & Judge, T. A. (1995). Measuring trust between organizational boundary role persons. *Organizational Behavior and Human Decision Processes*, *64*, 151-170.

Deutsch, M. (1958). Trust and suspicion. *Journal of Conflict Resolution*, *2*, 265-279.

Dirks, K. (1999). The effects of interpersonal trust on work group performance. *Journal of Applied Psychology*, *84*, 445-455.

Dirks, K., & Ferrin, D. L. (2002). Trust in leadership: Meta-analytic findings and implications for research and practice. *Journal of Applied Psychology*, *87*, 611-628.

Ferrin, D. L., & Dirks, K. T. (2003). The use of rewards to increase and decrease trust: Mediating processes and differential effects. *Organization Science*, *14*, 18-31.

Ferrin, D. L., Kim, P. H., Cooper, C. D., & Dirks, K. T. (2007). Silence speaks volumes: The effectiveness of reticence in comparison to apology and denial for responding to integrity- and competence-based trust violations. *Journal of Applied Psychology*, *92*, 893-908.

Hackman, R. J. (2003). *Leading teams*. Boston: Harvard Business School Publishing.

James, L., Demaree, R., & Wolf, G. (1984). Estimating within-group interrater reliability with and without response bias. *Journal of Applied Psychology*, *69*, 85-98.

Jarvenpaa, S., & Leidner, D. (1999). Communication and trust in global virtual teams. *Organization Science*, *10*, 791-815.

Johnson, D., & Grayson, K. (2005). Cognitive and affective trust in service delivery. *Journal of Business Research*, *58*, 500-507.

Jones, G. R., & George, J. M. (1998). The experience and evolution of trust: Implications for cooperation and teamwork. *Academy of Management Review*, *23*, 531-546.

Kim, P. H., Dirks, K. T., Cooper, C. D., & Ferrin, D. L. (2006). When more blame is better than less: The implications of internal vs. external attributions for the repair of trust after a competence- vs. integrity-based trust violation. *Organizational Behavior and Human Decision Processes*, *99*, 49-65.

Kim, P. H., Ferrin, D. L., Cooper, C. D., & Dirks, K. T. (2004). Removing the shadow of suspicion: The effects of apology versus denial for repairing competence- versus integrity-based trust violations. *Journal of Applied Psychology*, *89*, 104-118.

Kirkman, B. L., Rosen, B., Tesluck, P. E., & Gibson, C. B. (2006). Enhancing the transfer of computer assisted training proficiency in geographically distributed teams. *Journal of Applied Psychology*, *91*, 706-716.

Krishnan, R., & Martin, X. (2006). When does trust matter to alliance performance? *Academy of Management Journal*, *49*, 894-917.

Langfred, C. W. (2007) The downside of self-management: A longitudinal study of the effects of conflict on trust, autonomy, and task interdependence in self-managing teams. *Academy of Management Journal*, *50*, 885-900.

Lewicki, R. J., & Bunker, B. B. (1995). Developing and maintaining trust in work relationships. In R. M. Kramer & T. R. Tyler (Eds.), *Trust in organizations: Frontiers of theory and research* (pp. 114-139). Thousand Oaks, CA: Sage.

Lewis, J. D., & Weingert, A. (1985). Trust as a social reality. *Social Forces*, *63*, 967-985.

Mayer, R. C., & Davis, J. H. (1999). The effect of the performance appraisal system on trust for management: A field quasi-experiment. *Journal of Applied Psychology*, *84*, 123-136.

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrated model of organizational trust. *Academy of Management Review*, *20*, 709-739.
Mayer, R. C., & Gavin, M. B. (2005). Trust in management and performance: Who minds the shop while the employees watch the boss? *Academy of Management Journal*, *48*, 874-888.
Mayerson, D., Weick, K. E., & Kramer, R. M. (1996). Early trust and temporary groups. In R. M. Kramer & T. R. Tyler (Eds.), *Trust in organizations: Frontiers of theory and research* (pp. 166-195). Thousand Oaks, CA: Sage.
McAllister, D. J. (1995). Affect- and cognitive-based trust as foundations for interpersonal cooperation in organizations. *Academy of Management Journal*, *38*, 24-59.
McKnight, D. H., Cummings, L. L., & Chervany, N. L. (1998). Early trust formations in new organizational relationships. *Academy of Management Review*, *23*, 473-490.
Organ, D. W. (1988). *Organizational citizenship behavior: The "good soldier" syndrome*. Lexington, MA: Lexington Books.
Polzner, J. T., Crisp, C. B., Jarvenpaa, S. L., & Kim, J. W. (2006). Extending the faultline model to geographically dispersed teams: How collocated subgroups can impair group functioning. *Academy of Management Journal*, *49*, 679-692.
Prichard, J. S., & Ashleigh, M. J. (2007). The effects of team-skills training on transactive memory and performance. *Small Group Research*, *38*, 696-726.
Rempel, J. K., Holmes, J. G., & Zanna, M. P. (1985). Trust in close relationships. *Journal of Personality and Social Psychology*, *49*, 95-112.
Ring, S. M., & Van de Ven, A. (1992). Structuring cooperative relationships between organizations. *Strategic Management Journal*, *13*, 483-498.
Rousseau, D. M., Sitkin, S. B., Burt, R. S., & Camerer, C. 1998. Not so different after all: A cross-discipline view of trust. *Academy of Management Review*, *23*, 393-404.
Scanzoni, J. (1979). Social exchange and behavioural interdependence. In R. L. Burgess & T. L. Huston (Eds.), *Social exchange in developing relationships* (pp. 61-98). New York: Academic Press.
Schoorman, F. D., Mayer, R. C., & Davis, J. H. (1996, April). *Empowerment in veterinary clinics: The role of trust in delegation*. Paper presented at the 11th annual meeting of the Society for Industrial and Organizational Psychology, San Diego, CA.
Shapiro, D. L., Sheppard, B. H., & Cheraskin, L. (1992). Business on a handshake. *Negotiation Journal*, *8*, 365-377.
Sitkin, S. B., & Roth, N. L. (1993). Explaining the limited effectiveness of legalistic "remedies" for trust/distrust. *Organization Science*, *4*, 367-392.
Smith, C. A., Organ, D. W., & Near, J. P. (1983). Organizational citizenship behavior: Its nature and antecedents. *Journal of Applied Psychology*, *68*, 653-663.
Surva, M. A., Fuller, M. A., & Mayer, R. C. (2005). The reciprocal nature of trust: A longitudinal study of interacting teams. *Journal of Organizational Behavior*, *26*, 625-648.
Webber, S. S., Chen, G., Marsh, S., & Payne, S. (1999). Performance appraisal research contributes to team mental model measurement. In S. J. Zaccaro (Chair) & D. A. Hofmann (Discussant), B*reaking barriers in team research: New methodologies and applications for team research and practice*. Symposium conducted at the 1998 annual meeting of the Academy of Management, San Diego, CA.
Webber, S. S., & Klimoski, R. J. (2004). Client-project manager engagements, trust, and loyalty. *Journal of Organizational Behavior*, *25*, 997-1013.
Williams, M. (2001). In whom we trust: Group membership as an affective context for trust development. *Academy of Management Review*, *26*, 377-396.

Zaheer, A., McEvily, B., & Perrone, V. (1998). Does trust matter? Exploring the effects of interorganizational and interpersonal trust on performance. *Organization Science*, *9*, 141-159.

Zucker, L. G. (1986). Production of trust: Institutional sources of economic structure, 1840-1920. In B. M. Staw & L. L. Cummings (Eds.), *Research in organizational behavior* (pp. 53-111). Greenwich, CT: JAI.

**Sheila Simsarian Webber** (PhD, George Mason University) is an associate professor of management in the Sawyer Business School at Suffolk University. Her research focuses on team composition, trust, and project manager–client engagements.